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PalmSource
Conference 2000



Adding Communications to Your Applications

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About Mark/Space

- Founded October 31, 1990 (spooky!)
- Developing for Palm OS® since 1997
- Develop for most platforms
  - Mac, Windows and Palm OS® Platform
  - Software/Firmware for several SpringBoard™ Modules
  - Did Newton apps in a former life
- Came from Commercial/Shrink-wrap background

About Mark/Space (Cont.)

- We make communications software
  - PageNOW!—1- and 2-way Paging/SMS
  - Online—Terminal Emulation/Serial Debugging
  - PhoneWatcher—Caller ID/Call Tracking/Mobile Phone Book Editor
  - Fax—Send/Receive
  - Mac USB driver for Sony Clié
- And more recently, hardware…
  - DataCord—Serial data cables for Palm OS® devices, including Handspring Visor

Session Agenda

- Available Methods
- Using C++ as an Abstraction Tool
- Working with New Serial Manager
- Debugging Techniques
- Related Sessions
- Post-Conference Resources

Available Methods

- Exchange Manager (Beaming)
  - High level way to exchange typed data
- Internet Library (Wireless)
  - Ideal for wireless access to HTTP oriented data
- Net Library (TCP)
  - When you need raw, unlimited access to TCP oriented data, especially protocols other than HTTP—SMTP, POP3, Telnet, SNPP, etc.
- Serial Manager
  - Low level way to talk to devices in a byte oriented way over wire or wireless (IRComm, RFComm)

Exchange Manager (Beaming)

- Available since OS 3.0/IR devices
- Traditionally for exchange of typed data such as business cards, text memos, and other database records via IR
- Also works great for transaction oriented applications, such as turn-oriented games, PayPal, etc.
- Expanded in new OS

INetLib (Wireless)

- Available since Palm™ VII handheld
- Now expanded to Palm™ V and Handspring handhelds via OmniSky
- Also many flashable Palm OS® handhelds via Mobile Internet Kit
- Advantages
  - Hardware/Software Integration (some models)
- Disadvantages
  - Not a full TCP connection
  - Limited to HTTP put/get type operations

Netlib (TCP)

- Available since Pilot Professional (OS 2.0)
- Can use via many methods
  - Dialup via modem or mobile phone
  - Palm HotSync® Ethernet Cradle
  - MochaPPP and serial cradle/cable to a PC
  - Minstrel and other wireless modems
- Can code direct to Palm OS® API, or Berkeley Sockets

Serial Manager

- Available on all devices since day 1
- Can use via many methods
  - Dialup via modem or mobile phone
  - OmniSky
  - HotSync® Ethernet Cradle
  - MochaPPP and serial cradle/cable to a PC

Using C++ to Abstract Comms

- Provides a single API that you code your own apps to, hiding details of OS version, etc.
- Can use to handle many transports
  - Serial/Modem/RFComm (Bluetooth)
  - TCP/Telnet
- Caveats—the usual C++ with Palm OS® ones:
  - Can't call virtual methods unless globals available



The CConnection Base Class

- Low Level
  - Read/Write
  - Open/Close
  - Flush Buffer
  - IsConnected
  - ClearErr
- High Level
  - ExecuteCommand
  - GetResponse

Examples of Methods Specific to Sub Classes

- Serial
  - Get number of ports
  - Enumerate port names
  - Get port characteristics
- Modem
  - UI to ask for phone number
  - Extra code in close to hang up modem

New vs. Old Serial Manager

- Old (Classic) Serial Manager
  - Available on all devices
  - Your only choice for devices prior to OS 3.3
- New Serial Manager
  - More robust and complete calls
  - Allows for concept of multiple ports
  - Allows for plug-in drivers to be written

Example: Testing for New Serial Manager

```
Boolean NewSerialManagerPresent(void)
{
   Boolean present = false;
   Err err; UInt32 value;

   err = FtrGet(
      sysFileCSerialMgr,
      sysFtrNewSerialPresent, &value);

   present = ((err == 0) && (value != 0));

   return present;
}
```

Example: Unified API for Old/New Serial Manager

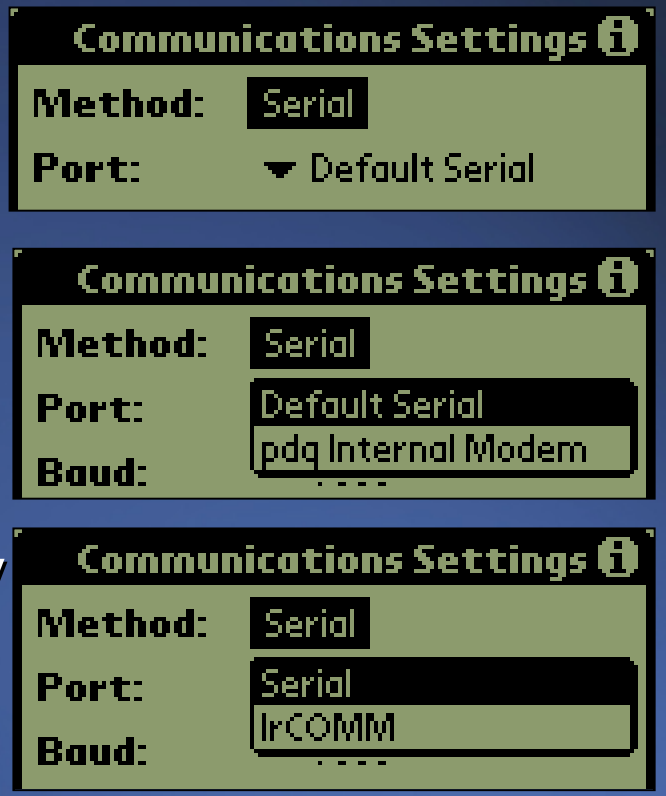
```
Err CSerialConnection::Read(MemPtr bufP,
UInt32 bufLen, Int32 timeout)
{
   Err  err = 0;
   UInt32  numReceived;

   if (fhasNewSerialMgr) {
      numReceived = SrmReceive(...
   } else {
      numReceived = SerReceive(...
   }

   return err;
}
```

Presenting a Port List

- For older OS, just show a single port
  - Can special case devices such as PDQ that have libraries to access additional ports
- For New Serial Manager, use available APIs to display list of available ports



Example: Getting Port Count

```
UInt16 CSerialConnection::GetNumDevices()
{
   UInt16 numDevices; Err err;

   if (fHasNewSerialMgr) {
      err = SrmGetDeviceCount(&numDevices);
      if (err) numDevices = 0;
   } else {
      if (fHaspdqModem) numDevices = 2;
      else numDevices = 1;
   }

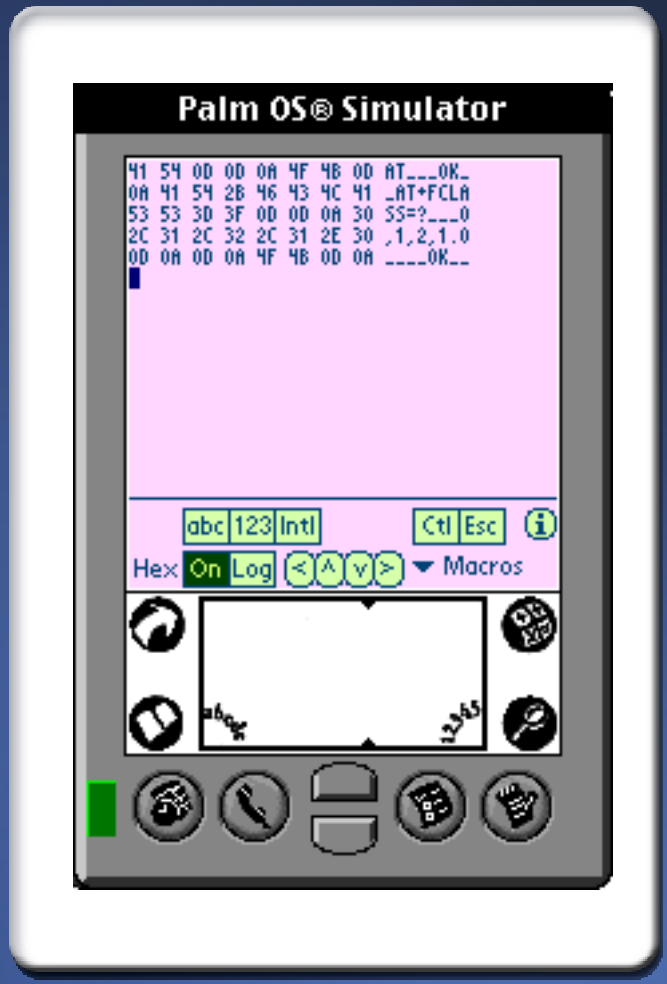
   return numDevices;
}
```

Debugging Techniques

- Simulator
- Emulator (POSE)
- Log Field
- Log File

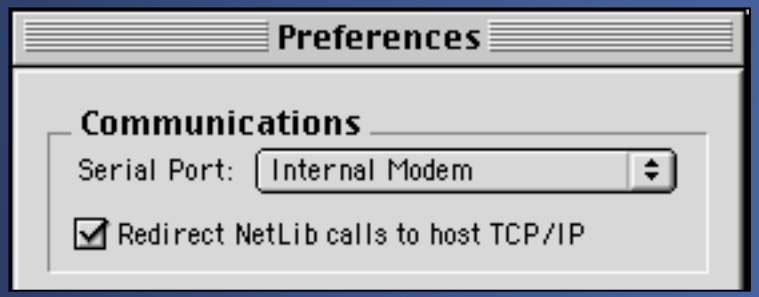
Debugging With Simulator

- Simulator
  - Mac only
  - Works best with 3.0 SDK
  - 3.5 SDK has, ahem, issues
  - Hopefully will be better going forward
  - My preferred tool when available



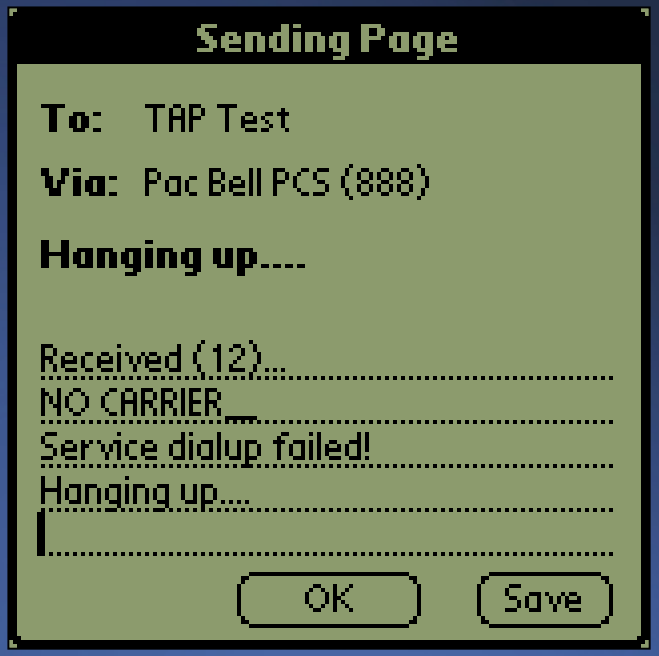
Debugging With POSE

- POSE
  - Multi-platform
  - Can emulate serial
    - And thus IRComm
      (if host PC/Mac has an IRComm virtual serial driver)
  - Can emulate TCP and INetLib/Palm.net®
  - Does not (currently) emulate pure IR



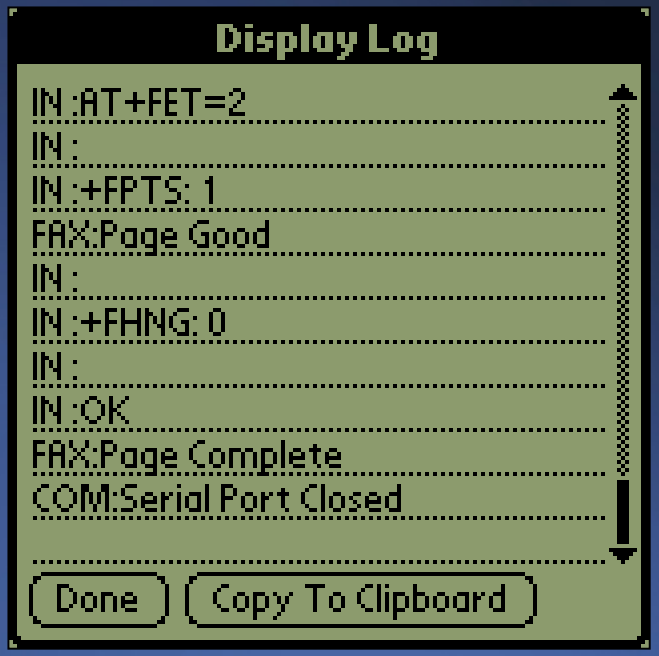
Debugging Techniques (Cont.)

- Log Field (Live)
  - Show data in a field as it is generated
  - Can also have “Copy to Clipboard” or “Save to Memo” for field testers to email back to you
  - Can provide a scrollbar and access to edit menu



Debugging Techniques (Cont.)

- Log “File” (post-mortem)
  - Could be to a private database or preference record
  - Or (to avoid viewer code) to a memo pad record
    - But then limited to 4K
- If not to memo pad, provide means to view the log after the fact



Selected War Stories/Gotchas

- Handspring serial port issues
  - Electrical
  - Port in use
    - Call HsExtKeyboardEnable(false)
- Memory allocation and SerSetReceiveBuffer
- False positive with SrmReceiveCheck
- Auto Off
- Software flow control not implemented

Conclusions

- There are many methods available to you
- Choose the one that makes sense for your app
- You can support more than one transport in the same app
- Debugging doesn't have to be (too) painful
- Most Gotchas discussed at one point or another on one of the mailing lists
  - Make use of the archives!
  - If that fails, post…

Related Sessions

- Prior (Video or Tape available)
  - 204 “Web Clipping APIs”
  - 205 “Bluetooth APIs”
  - 307 “Adding Wireless Capability to a PRC”
- Still remaining
  - 503 “Optimizing Use of the Serial Manager”
  - 109L “Debugging Communications”

Resources

- Technical
  - Palm, Inc. Mailing Lists/Newsgroups
    - palm-dev-forum@news.palm.com
    - comm-dev-forum@news.palmos.com
    - pqa-dev-forum@news.palmos.com
    - Join via http://www.palmos.com/dev/tech/support/forums/
  - Other Newsgroups
    - news://news.massena.com
- Marketing
  - Palm OS® Entrepreneurs Forum (Mailing List) www.markspace.com/pef/


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